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ARTIFICIAL INTELLIGENCE AND DATA ENGINEERING

Quantum Computing and Quantum Internet

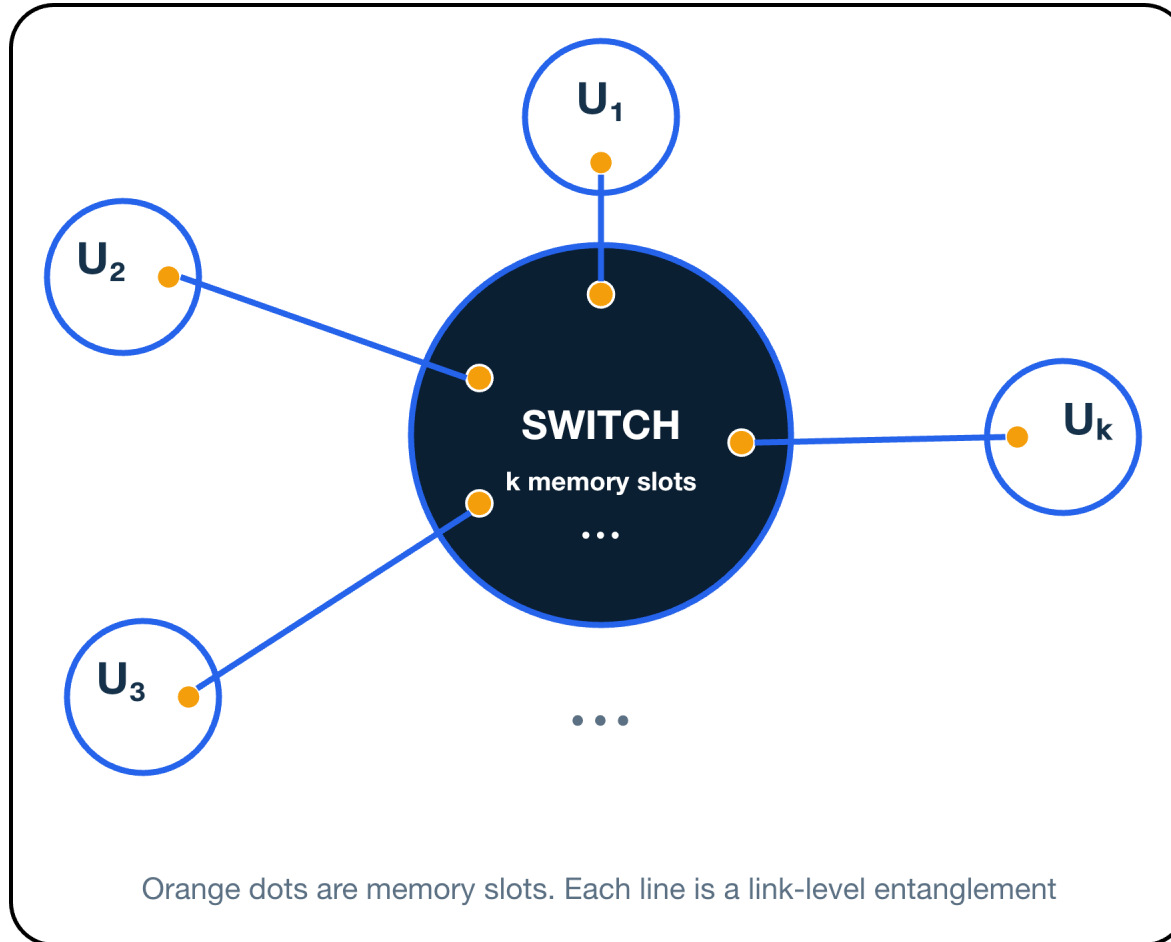
Entanglement Switch

End-to-end entanglement in a star network

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Star-network architecture

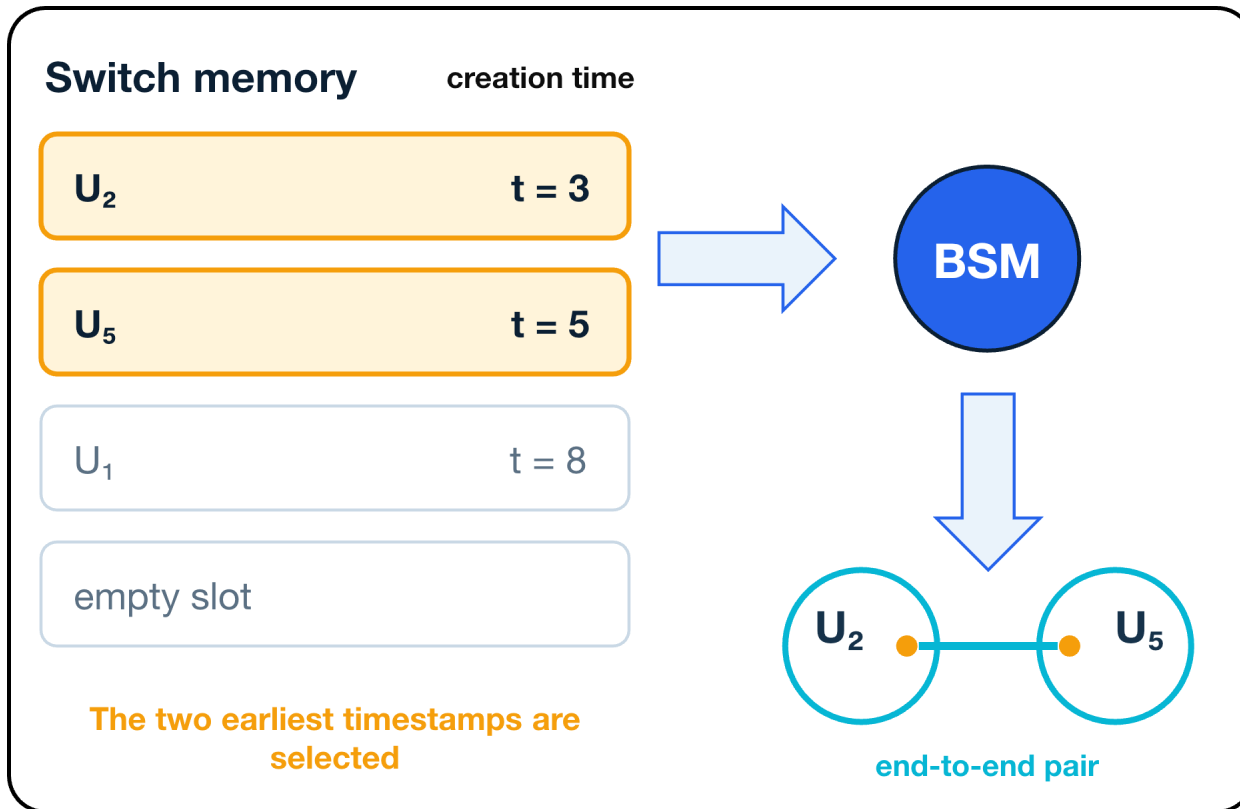


- k indistinguishable user nodes surround one central entanglement switch.
- Each user stores one qubit. The switch stores k qubits, one slot per link.
- Every idle link independently attempts to generate a link-level Bell pair.
- Two stored link-level pairs are converted into one end-to-end pair.

Objective: establish end-to-end entanglement between any two users.

Oldest-first switching policy

Users do not request a destination. Any two users can receive the next end-to-end pair.



- When at least two link-level pairs are stored, the switch selects the two oldest pairs.
- The switch performs an instantaneous Bell-state measurement and sends the result to both users.
- The users apply the required Pauli correction. The delivered pair is consumed immediately.
- The switch releases both slots, restarts generation on those links, and repeats while two pairs remain.

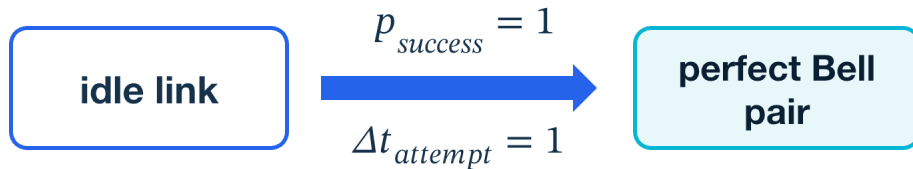
Ideal model assumptions

The ideal benchmark removes state imperfections and memory degradation.

Perfect link state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

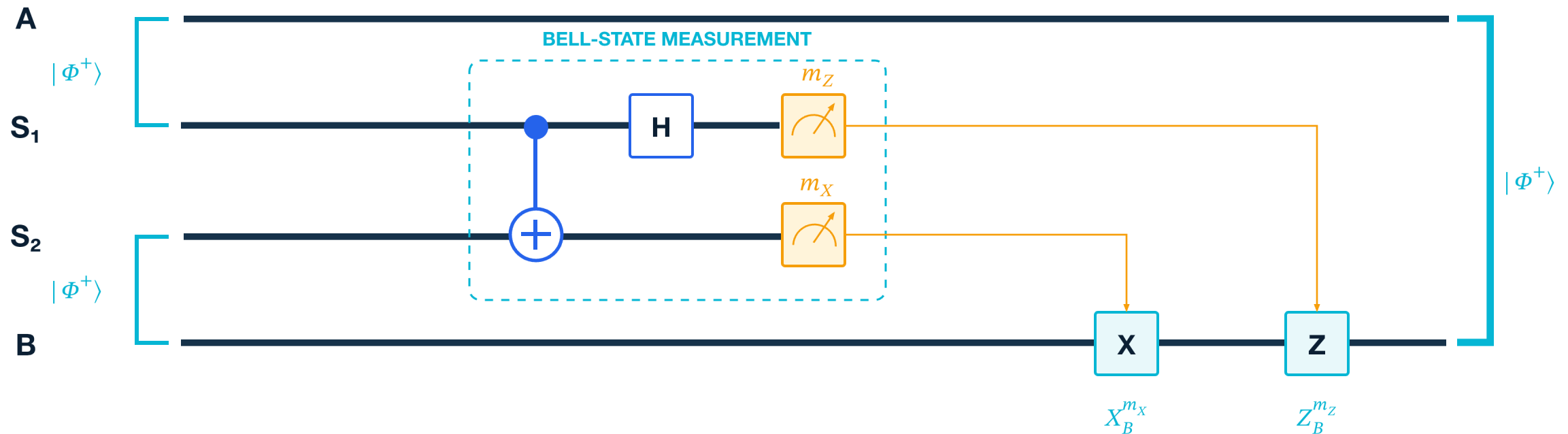
$$\rho_{link} = |\Phi^+\rangle\langle\Phi^+|$$



- Every generation attempt succeeds with certainty.
- Each attempt lasts one discrete time unit
- Quantum memories are noiseless, so stored states do not decohere.
- Bell-state measurements and Pauli corrections are instantaneous.

Ideal entanglement-swapping circuit

$$|\Psi_0\rangle = |\Phi^+\rangle_{AS_1} \otimes |\Phi^+\rangle_{S_2B}$$



A, B are qubits inside the nodes and S₁, S₂ are qubits inside the switch

Ideal output and throughput

Output state after the BSM

$$\rho_{AB}^{out} = |\Phi^+\rangle_{AB}\langle\Phi^+|$$

$$\rho_B^{out} = \text{Tr}_A(\rho_{AB}^{out}) = \frac{I_B}{2}$$

Output Fidelity

$$F_{swap} = \langle\Phi^+|\rho_{AB}^{out}|\Phi^+\rangle = 1$$

Expected throughput

- Each end-to-end pair consumes two link-level pairs.
- The instantaneous BSM repeats until fewer than two pairs remain.

$$N_{swap} = \lfloor \frac{k}{2} \rfloor$$

$$T_{ideal} = \frac{N_{swap}}{\Delta t} = \lfloor \frac{k}{2} \rfloor$$

Non-ideal model assumptions

Link generation and memory storage are non-ideal; BSMs and Pauli corrections remain ideal and instantaneous.

Werner-state link generation

$$\rho_W(w) = w |\Phi^+\rangle\langle\Phi^+| + (1-w) \frac{I_4}{4}$$

The Werner parameter $w \in [0, 1]$ controls the Bell component.
The Fidelity is

$$F_{link} = \langle\Phi^+ | \rho_W(w) | \Phi^+\rangle = \frac{1+3w}{4}$$

Probabilistic pair generation

- Each independent Bernoulli attempt takes one discrete time unit.
- A failed link retries in the next available time step.

Memory depolarization

$$D_{p_w}(\rho) = p_w \rho + (1 - p_w) \text{Tr}(\rho) \frac{I_2}{2} \quad (1)$$

p_w is the one-step retained-state factor: under this convention, it is the probability of no depolarization event. The local channel acts independently on every memory.

QuantumSavory lifetime parameter

$$p_w = e^{-1/\tau} \quad (2)$$

$$\tau = -\frac{1}{\ln(p_w)}$$

$p_w = 1$ corresponds to $\tau = \infty$ and disables memory noise.

Two Werner pairs before swapping

This derivation isolates imperfect link states with memory noise disabled: $p_w = 1$

Link A-S₁

$$\rho_{AS_1}(w_1) = w_1 |\Phi^+\rangle_{AS_1} \langle \Phi^+| + (1 - w_1) \frac{I_{AS_1}}{4} \quad (3)$$

First independently generated link-level pair

Link S₂-B

$$\rho_{S_2B}(w_2) = w_2 |\Phi^+\rangle_{S_2B} \langle \Phi^+| + (1 - w_2) \frac{I_{S_2B}}{4} \quad (4)$$

Second independently generated link-level pair

Independent four-qubit input

$$\rho_0 = \rho_{AS_1}(w_1) \otimes \rho_{S_2B}(w_2) \quad (5)$$

The tensor product assumes that the two link-generation processes are independent.

Werner output after an ideal BSM

The ideal BSM and outcome-dependent Pauli correction preserve the Werner form.

End-to-end Werner state

$$\rho_{AB}^{out} = w_1 w_2 |\Phi^+\rangle_{AB} \langle \Phi^+| + (1 - w_1 w_2) \frac{I_{AB}}{4} \quad (6)$$

$$\rho_{AB}^{out} = \rho_W(w_1 w_2), w_{swap} = w_1 w_2$$

Delivered fidelity

$$\begin{aligned} F_{swap} &= \langle \Phi^+ | \rho_{AB}^{out} | \Phi^+ \rangle \\ &= \frac{1 + 3w_1 w_2}{4} \quad (7) \end{aligned}$$

SIMULATED SYMMETRIC CASE

$$w_1 = w_2 = w \Rightarrow F_{swap} = \frac{1 + 3w^2}{4}$$

Keeping w_1 and w_2 distinct also covers unequal effective link qualities after different storage times.

Depolarization of a Werner pair

Memory depolarization rescales the Pauli correlations while preserving the Werner form.

Write the Werner state in the Pauli basis:

$$\rho_W(w) = \frac{1}{4} [I \otimes I + w(X \otimes X - Y \otimes Y + Z \otimes Z)] \quad (8)$$

Over one time unit, memory depolarization preserves the identity and contracts each Pauli operator:

$$\begin{aligned} D_{p_w}(I) &= I & D_{p_w}(X) &= p_w X \\ D_{p_w}(Y) &= p_w Y & D_{p_w}(Z) &= p_w Z \end{aligned} \quad (9)$$

The channel acts on both memories for time t , giving two retention factors per correlation:

$$\begin{aligned} & (D_{p_w}^t \otimes D_{p_w}^t)[\rho_W(w)] \\ &= \frac{1}{4} [I \otimes I + w p_w^{2t} (X \otimes X - Y \otimes Y + Z \otimes Z)] \\ &= \rho_W(w p_w^{2t}) \end{aligned} \quad (10)$$

Final swapping fidelity with memory noise

Different storage times give the two link pairs different effective Werner parameters.

Let t_i be the age of link pair i when the BSM starts. Both stored qubits experience the local channel for this time:

$$w_1^{eff} = w p_w^{2t_1} \qquad w_2^{eff} = w p_w^{2t_2}$$

Insert these effective parameters into the Werner-only swapping result from Eq. (7):

$$F_{swap} = \frac{1 + 3w_1^{eff} w_2^{eff}}{4}$$

The two effective parameters multiply:

$$w_1^{eff} w_2^{eff} = w^2 p_w^{2t_1} p_w^{2t_2} = w^2 p_w^{2(t_1 + t_2)}$$

Therefore, the delivered fidelity is

$$F_{swap}(t_1, t_2) = \frac{1 + 3w^2 p_w^{2(t_1 + t_2)}}{4} \tag{11}$$

With memory noise, longer storage lowers fidelity toward 1/4. Link success affects fidelity through the storage times t_1 and t_2 .

Performance metrics

FIDELITY

$$F_j = \langle \Phi^+ | \rho_{uv}^{(j)} | \Phi^+ \rangle$$

Quality of the two-user state relative to the ideal Bell state. It is evaluated after the Pauli correction determined by the BSM outcome.

WAITING TIME

$$W_j = t_{BSM,j} - \min(t_{1,j}, t_{2,j})$$

Time spent in memory by the older of the two link-level pairs selected for the swap, measured from its creation to the BSM.

THROUGHPUT

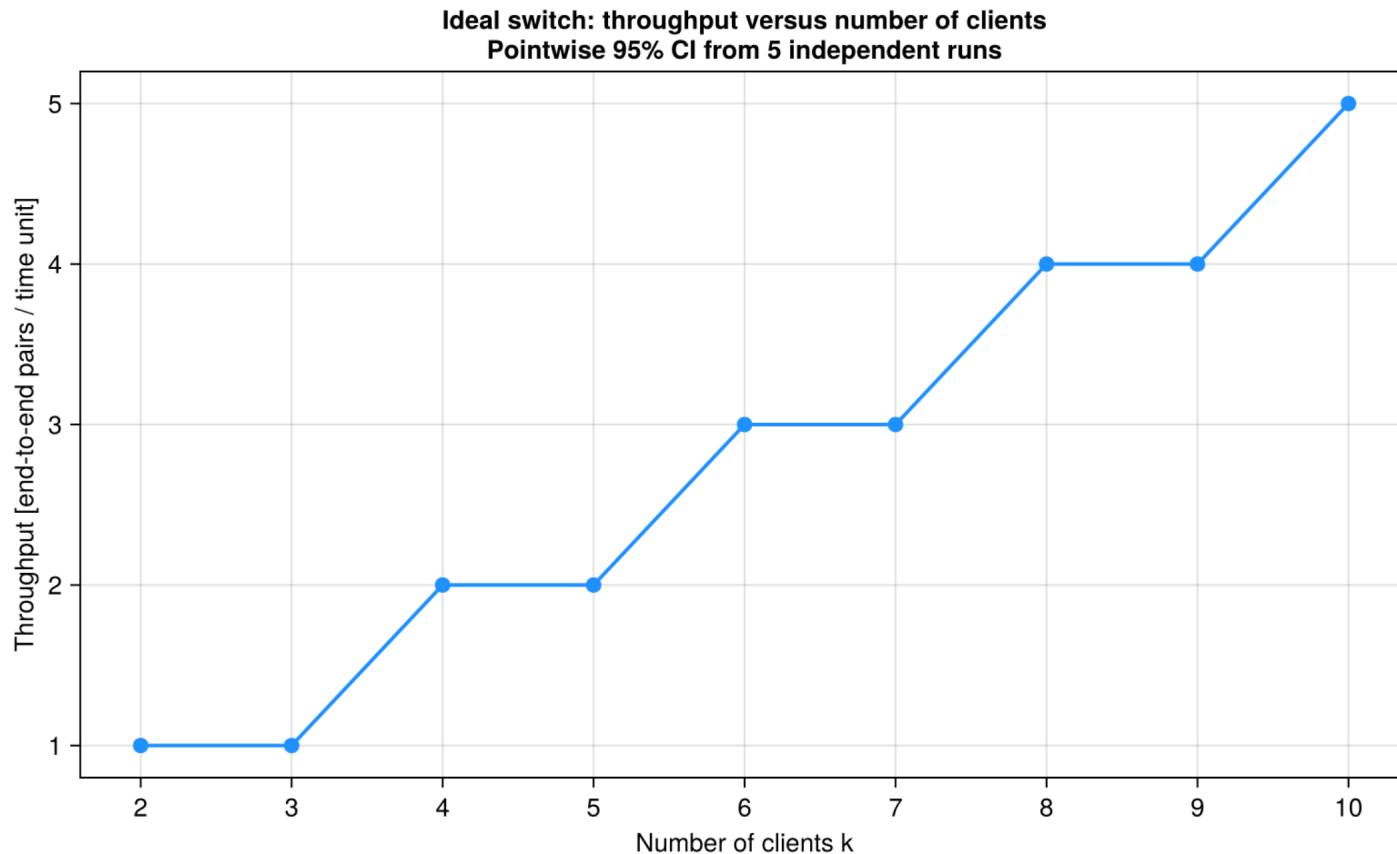
$$R = \frac{N_{delivered}}{\Delta t_{steady}}$$

Average number of end-to-end pairs delivered per time unit, evaluated in steady state over a sufficiently long simulation horizon.

Ideal throughput matches the pairing limit

$k = 2-10$ · $p_{\text{success}} = 1$ · perfect Bell pairs · noiseless memories

Five runs per k , ending at $T = 101$. Throughput uses the final 100 time units.



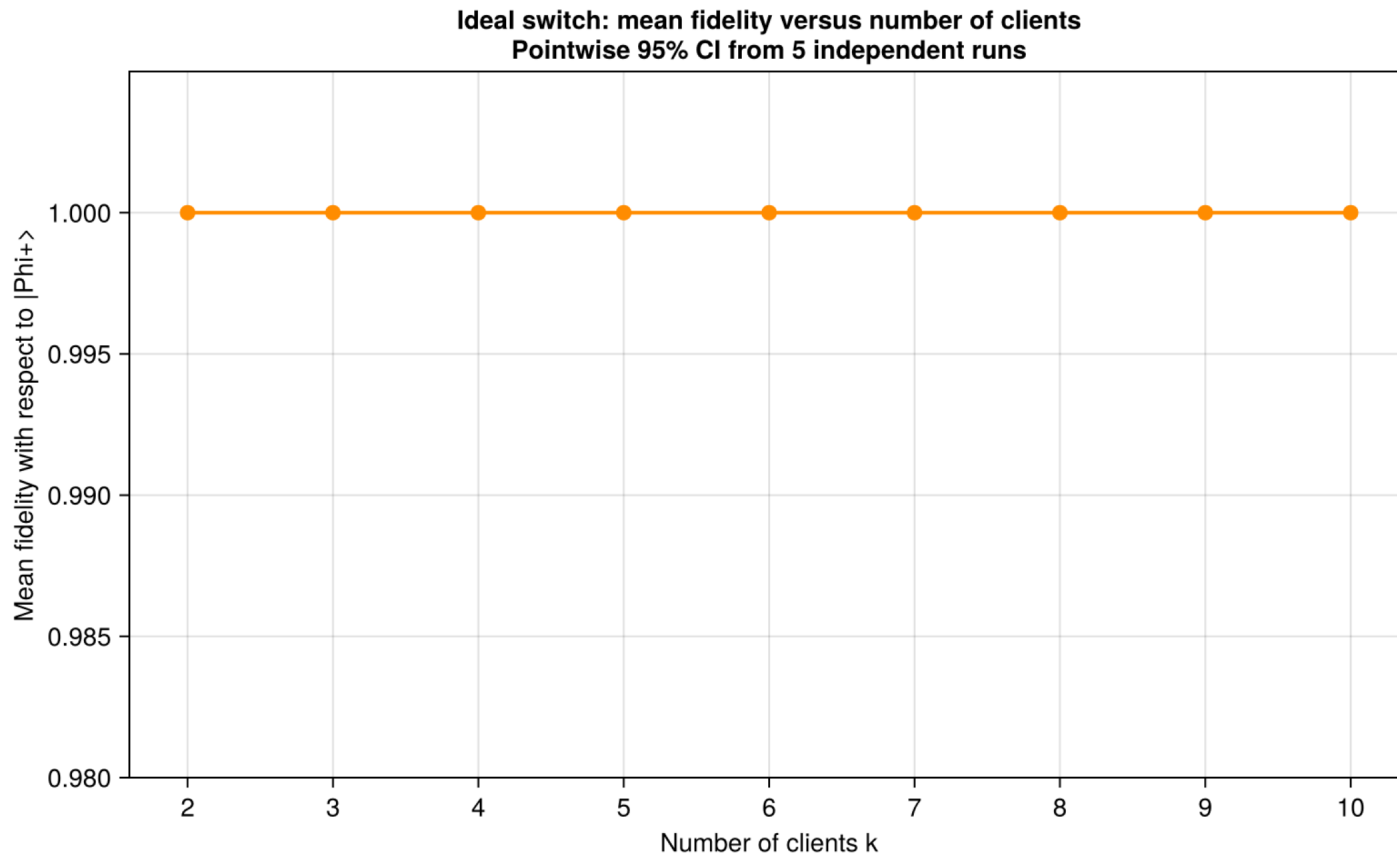
INTERPRETATION

Each delivered pair consumes two link-level Bell pairs, giving $\lfloor k/2 \rfloor$ deliveries per time unit. For odd k , one pair stays in memory and occupies its link slot. Only $k-1$ links generate in the next round. This explains the plateaus at $k = 2-3$, $4-5$, $6-7$ and $8-9$. Adding one client does not always increase the rate.

Ideal fidelity remains equal to one

$k = 2-10$ · $p_{\text{success}} = 1$ · ideal BSM and Pauli corrections · no memory noise

Five runs per k , ending at $T = 101$. Every delivered pair contributes to the mean fidelity.



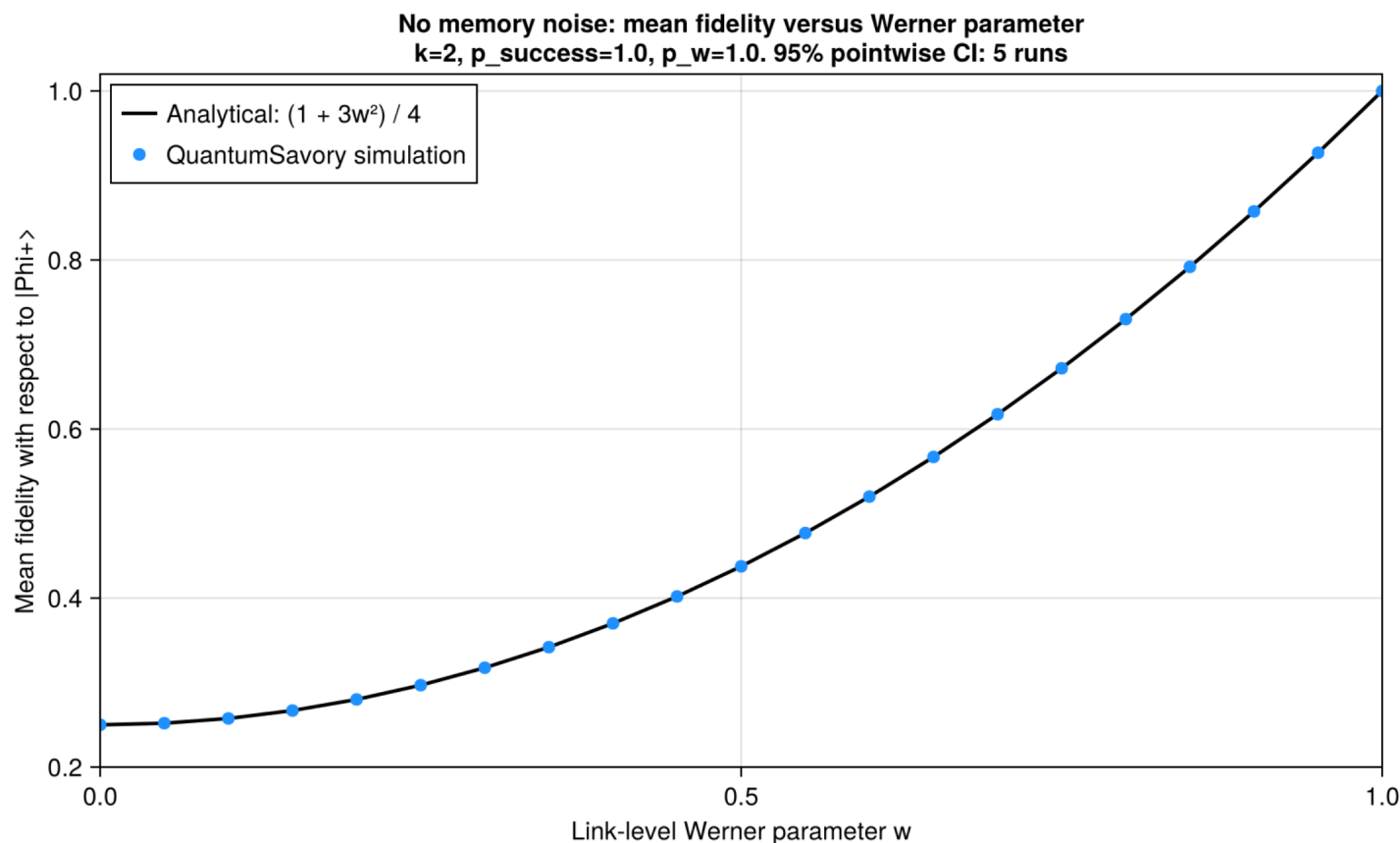
INTERPRETATION

Perfect input Bell pairs and the outcome-dependent Pauli correction deliver $|\Phi^{++}\rangle$ after every swap. Noiseless memories preserve the state even when a link pair waits for a partner. The flat $F = 1$ curve confirms that increasing k changes capacity without reducing fidelity in the ideal model.

Werner-state fidelity follows theory

$k = 2$ · $p_{\text{success}} = 1$ · $p_w = 1$ · w swept from 0 to 1

Five runs for each of 21 values of w , ending at $T = 21$. All delivered-pair fidelities are retained.



INTERPRETATION

Deterministic generation and noiseless memories isolate the quality of the input link states.

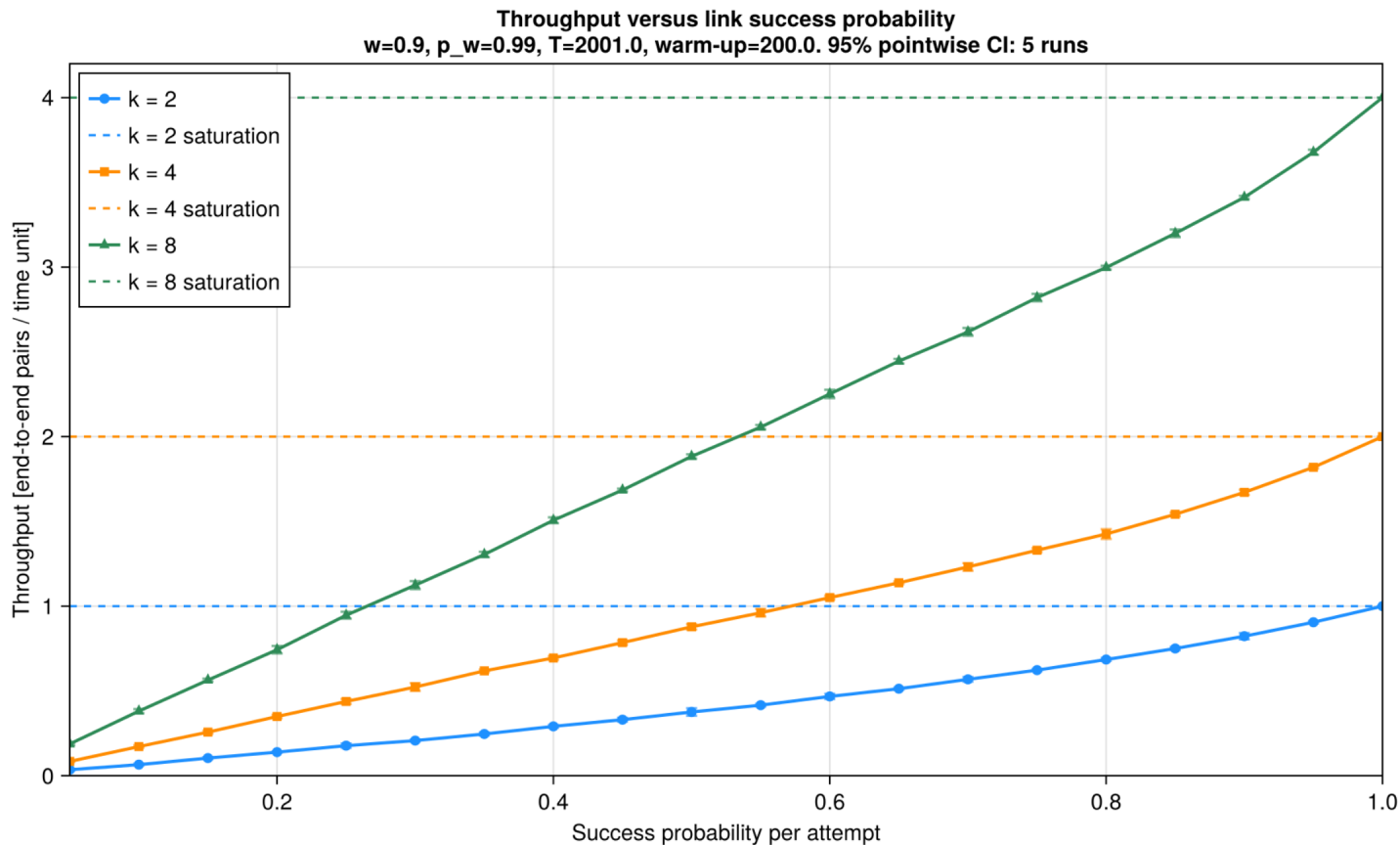
Swapping multiplies the two Werner parameters: $w_{\text{out}} = w^2$, so $F = (1 + 3w^2)/4$.

All 21 points follow the quadratic law, from $F = 1/4$ to $F = 1$. The largest discrepancy is 3.33×10^{-16} , consistent with round-off.

Link success probability controls throughput

$w = 0.9$ · $p_w = 0.99$ · $k = 2, 4, 8$ · p_{success} swept from 0.05 to 1

Five runs per point, ending at $T = 2001$. The first 200 time units are excluded from throughput.



INTERPRETATION

More successful link attempts supply pairs faster. Throughput rises approximately linearly over much of the tested range.

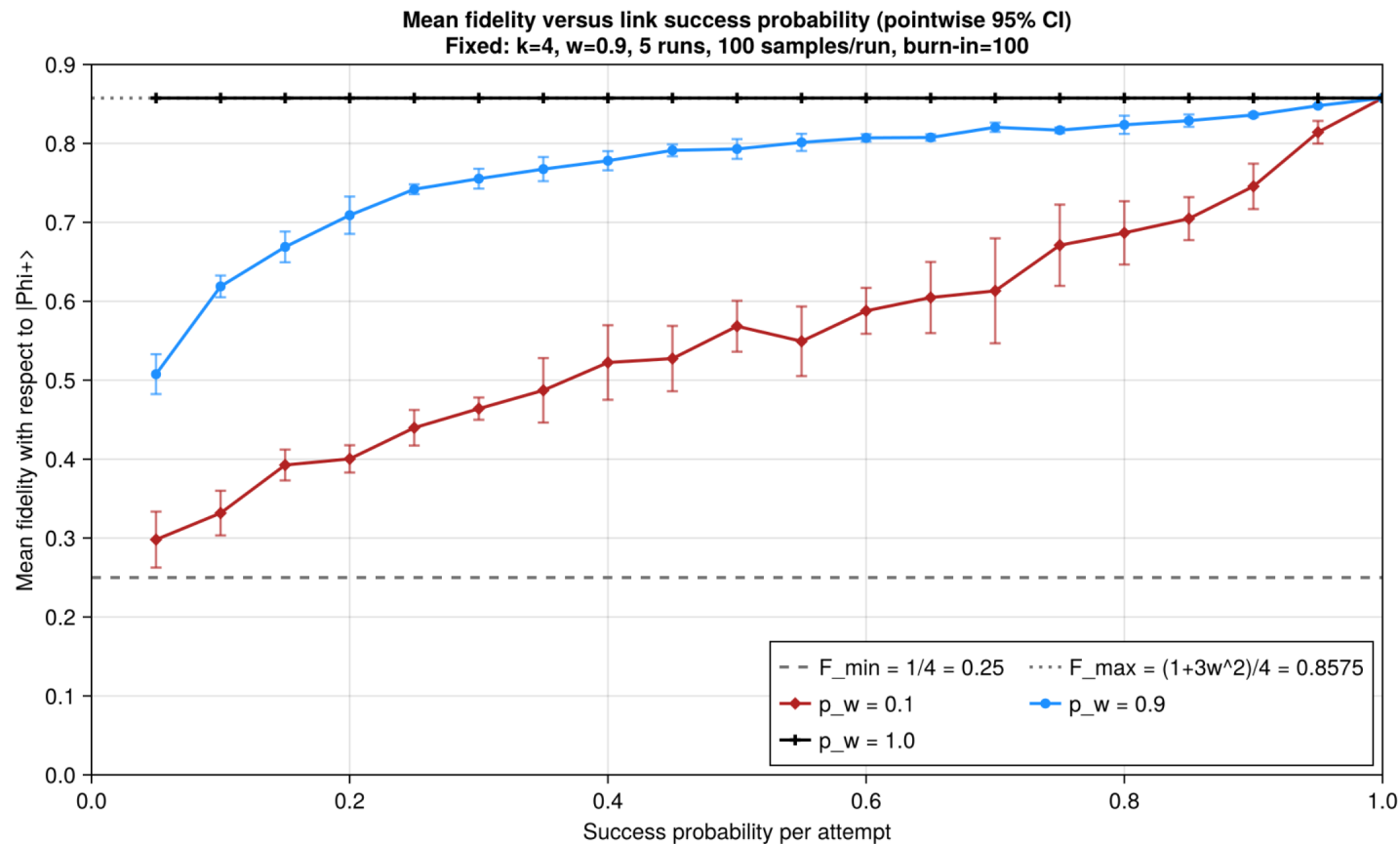
More clients add links that generate in parallel, increasing the total delivery rate.

At $p_{\text{success}} = 1$, $k = 2, 4$ and 8 reach 1, 2 and 4 pairs per time unit, exactly their $k/2$ saturation limits.

Faster link generation protects fidelity

$k = 4$ · $w = 0.9$ · $p_w = 0.1, 0.9, 1.0$ · p_{success} swept from 0.05 to 1

Five runs per point. Per run: discard the first 100 deliveries, then average the next 100.



INTERPRETATION

Higher link success shortens the wait for a second pair, leaving less time for depolarization before swapping.

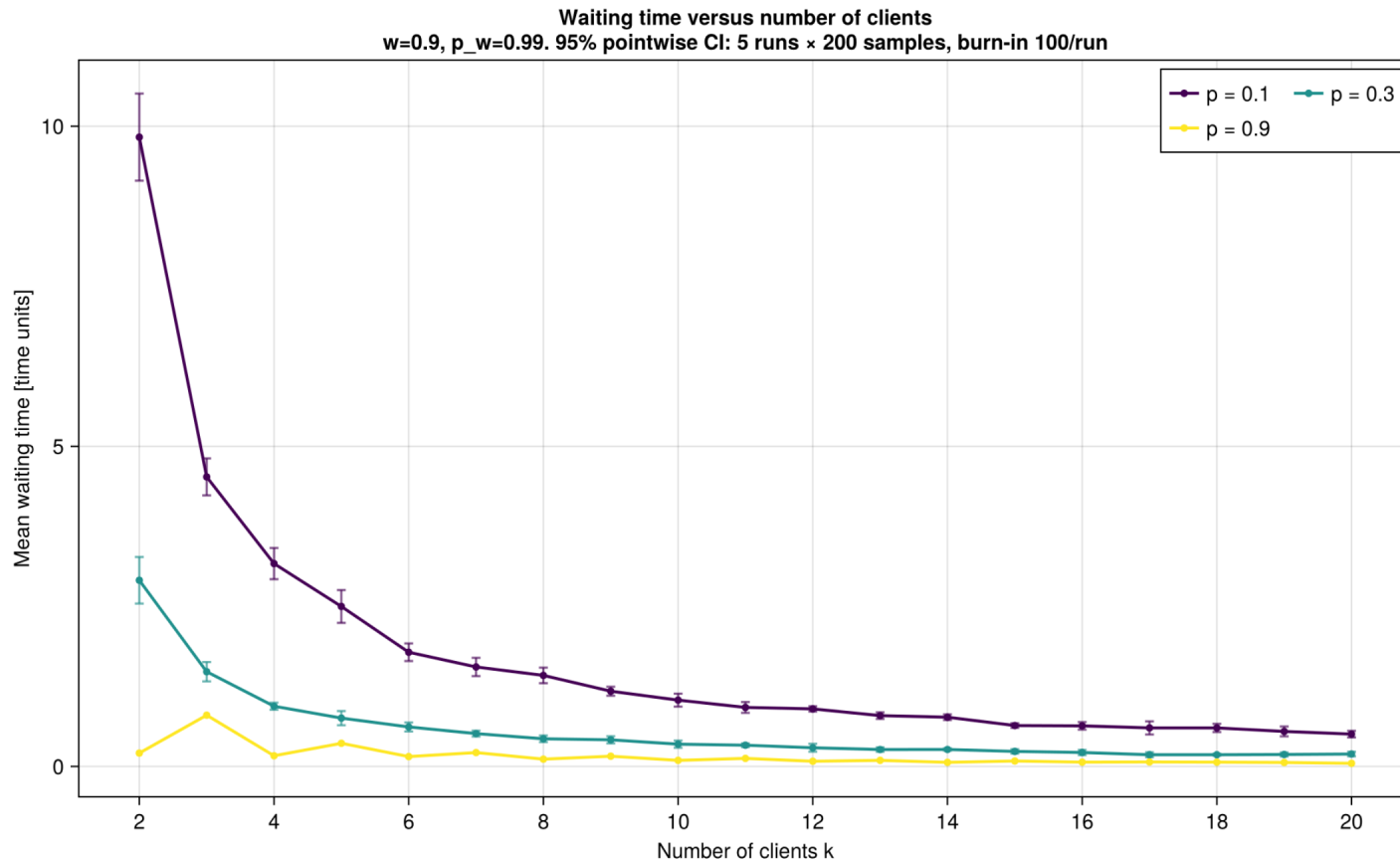
At $p_{\text{success}} = 1$, all curves reach $(1 + 3w^2)/4 = 0.8575$. With $p_w = 1$, fidelity stays at this value throughout.

At $p_w = 0.1$ and $p_{\text{success}} = 0.05$, $F \approx 0.2981$, close to the fully depolarized limit $1/4$.

More clients reduce pair waiting time

$w = 0.9$ · $p_w = 0.99$ · $p_{\text{success}} = 0.1, 0.3, 0.9$ · k swept from 2 to 20

Five runs per point. Per run: discard the first 100 swaps, then average the next 200 waiting times.



INTERPRETATION

More clients give a stored pair more chances to find a partner, so the mean wait generally decreases.

At $k = 2$, raising p_{success} from 0.1 to 0.9 reduces the mean wait from 9.829 to 0.208 time units.

At $p_{\text{success}} = 0.9$, raising k from 2 to 20 lowers the mean from 0.208 to 0.048 time units.

The graph also suggests an odd-even pattern. What happens is that for odd values of k there are more chances for a link-level pair to not find another pair for entanglement swapping, needing to wait more time. This is much more noticeable for low values of k .